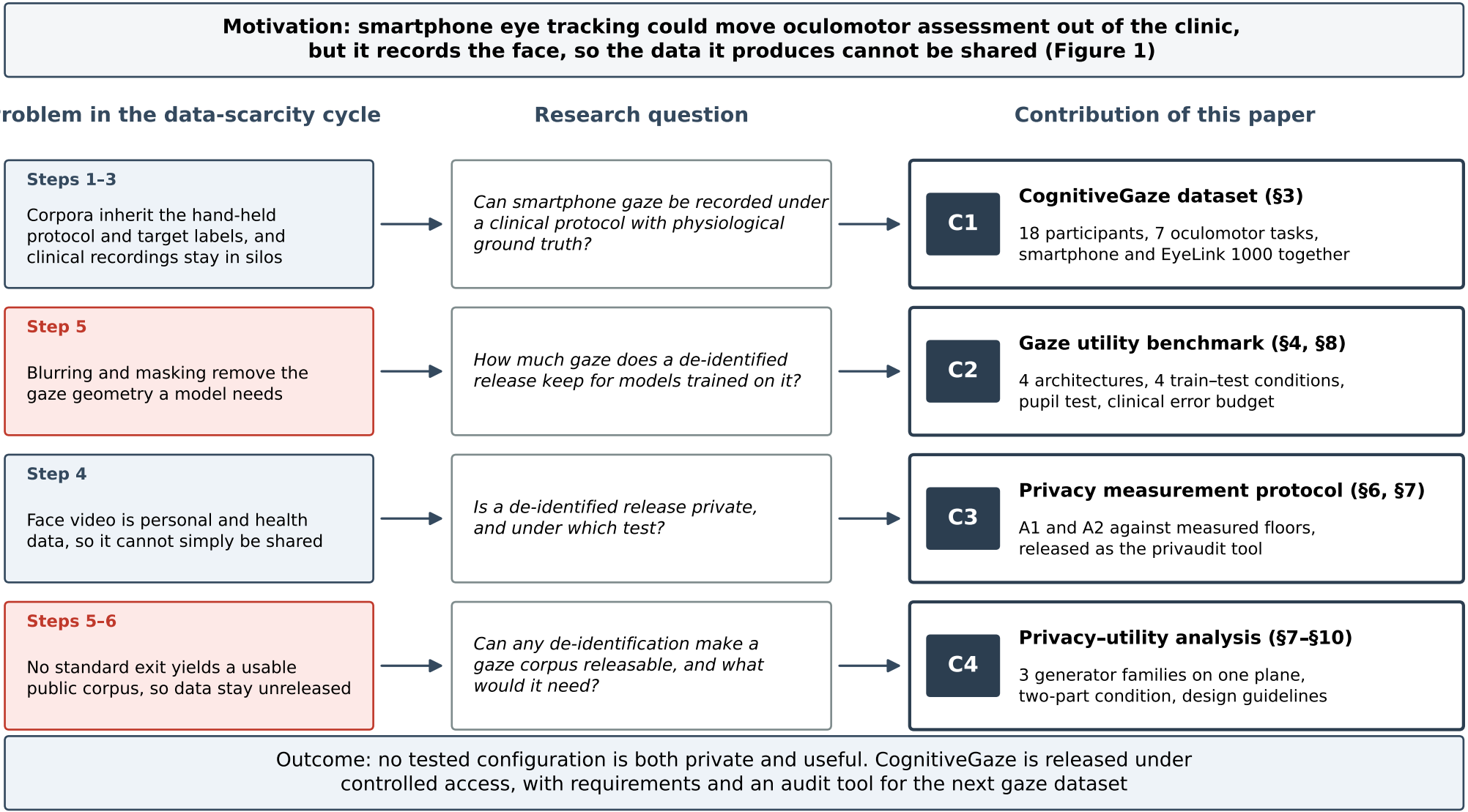
Motivation: smartphone eye tracking could move oculomotor assessment out of the clinic, but it records the face, so the data it produces cannot be shared (Figure 1)
Problem in the data-scarcity cycle
Research question
Contribution of this paper
Steps 1–3
Corpora inherit the hand-held protocol and target labels, and clinical recordings stay in silos
Can smartphone gaze be recorded under a clinical protocol with physiological ground truth?
C1
CognitiveGaze dataset (§3)
18 participants, 7 oculomotor tasks, smartphone and EyeLink 1000 together
Step 5
Blurring and masking remove the gaze geometry a model needs
How much gaze does a de-identified release keep for models trained on it?
C2
Gaze utility benchmark (§4, §8)
4 architectures, 4 train–test conditions, pupil test, clinical error budget
Step 4
Face video is personal and health data, so it cannot simply be shared
Is a de-identified release private, and under which test?
C3
Privacy measurement protocol (§6, §7)
A1 and A2 against measured floors, released as the privaudit tool
Steps 5–6
No standard exit yields a usable public corpus, so data stay unreleased
Can any de-identification make a gaze corpus releasable, and what would it need?
C4
Privacy–utility analysis (§7–§10)
3 generator families on one plane, two-part condition, design guidelines
Outcome: no tested configuration is both private and useful. CognitiveGaze is released under controlled access, with requirements and an audit tool for the next gaze dataset